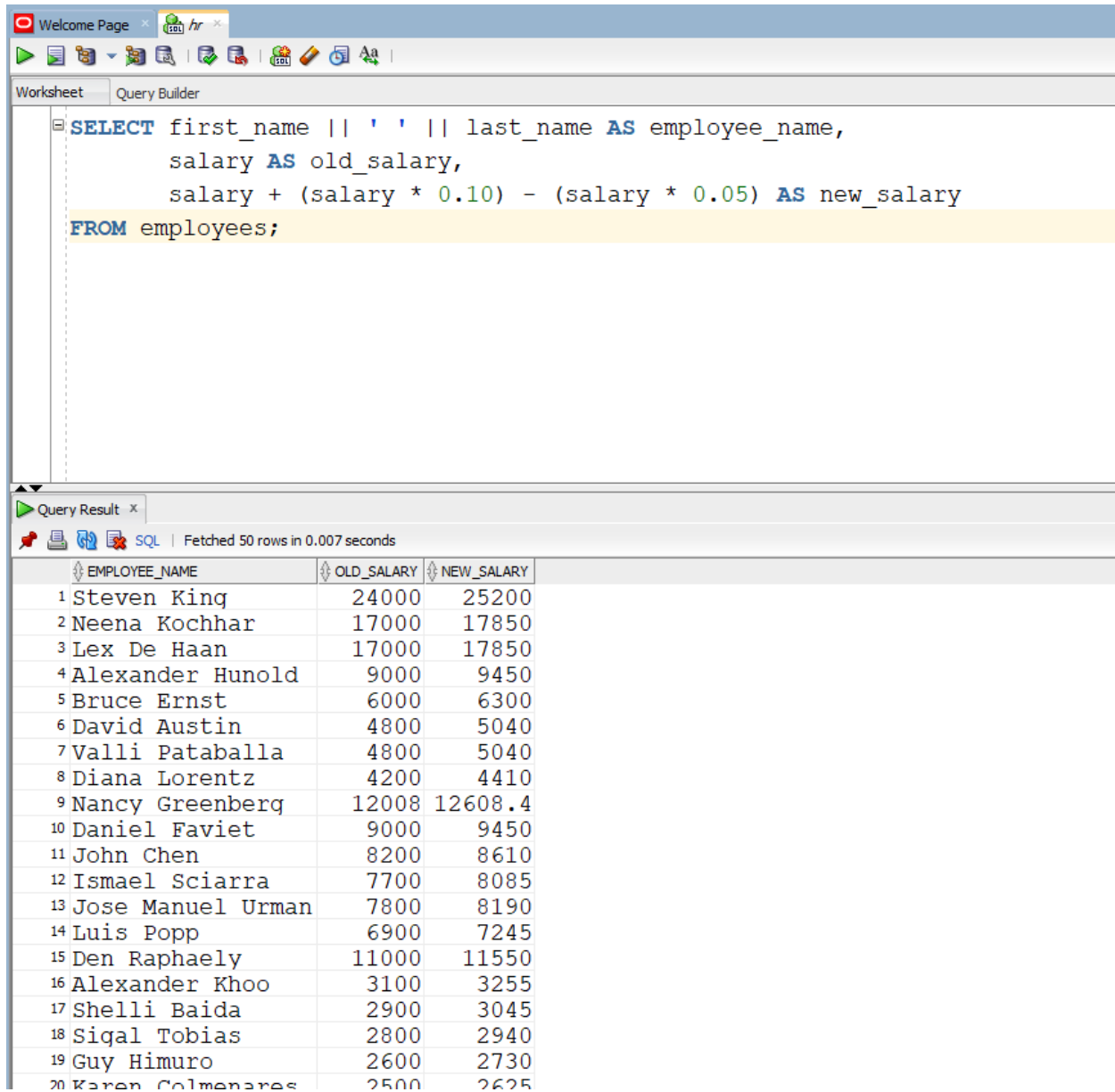


DBS LAB 4

IN-LAB TASKS:

Task 1



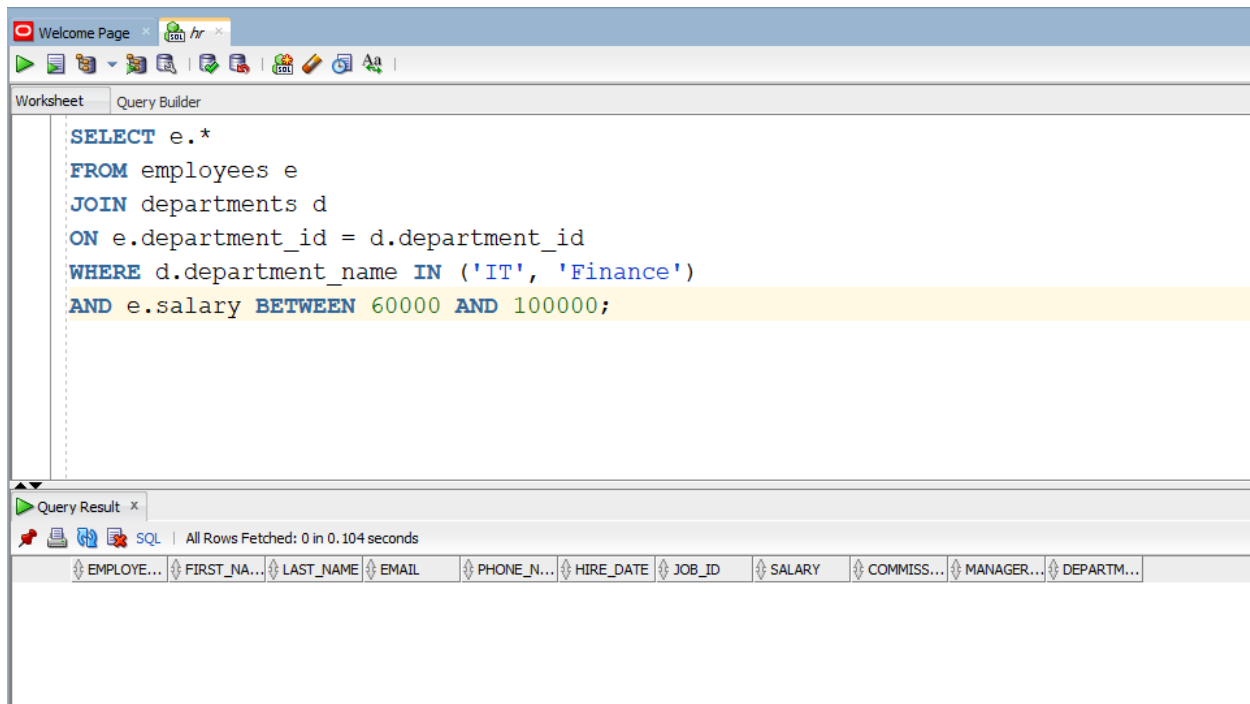
The screenshot shows a database query tool interface. The top part is the 'Query Builder' tab, which contains the following SQL query:

```
SELECT first_name || ' ' || last_name AS employee_name,  
       salary AS old_salary,  
       salary + (salary * 0.10) - (salary * 0.05) AS new_salary  
FROM employees;
```

The bottom part is the 'Query Result' tab, which displays the results of the query. It shows a table with three columns: EMPLOYEE_NAME, OLD_SALARY, and NEW_SALARY. The table contains 20 rows of data, representing employees and their salaries before and after a 5% increase.

	EMPLOYEE_NAME	OLD_SALARY	NEW_SALARY
1	Steven King	24000	25200
2	Neena Kochhar	17000	17850
3	Lex De Haan	17000	17850
4	Alexander Hunold	9000	9450
5	Bruce Ernst	6000	6300
6	David Austin	4800	5040
7	Valli Pataballa	4800	5040
8	Diana Lorentz	4200	4410
9	Nancy Greenberg	12008	12608.4
10	Daniel Faviat	9000	9450
11	John Chen	8200	8610
12	Ismael Sciarra	7700	8085
13	Jose Manuel Urman	7800	8190
14	Luis Popp	6900	7245
15	Den Raphaely	11000	11550
16	Alexander Khoo	3100	3255
17	Shelli Baida	2900	3045
18	Sigal Tobias	2800	2940
19	Guy Himuro	2600	2730
20	Karen Colmenares	2500	2625

Task 2

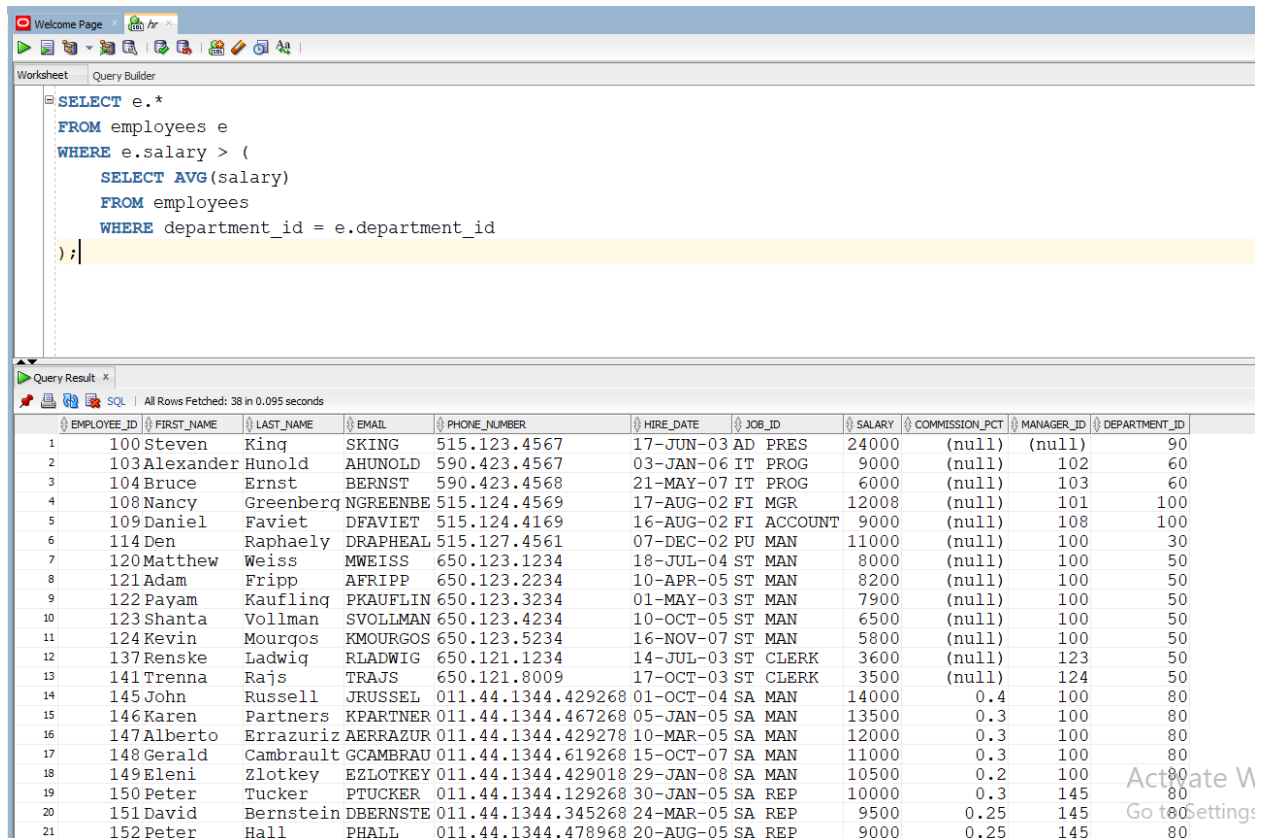


The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT e.*
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name IN ('IT', 'Finance')
AND e.salary BETWEEN 60000 AND 100000;
```

The 'Query Result' window below shows the results of the query. The status bar indicates 'All Rows Fetched: 0 in 0.104 seconds'. The column headers are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID.

Task 3



The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT e.*
FROM employees e
WHERE e.salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
);
```

The 'Query Result' window below shows the results of the query. The status bar indicates 'All Rows Fetched: 38 in 0.095 seconds'. The column headers are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID.

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	17-JUN-03	AD PRES	24000	(null)	(null)	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT PROG	9000	(null)	102	60
104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT PROG	6000	(null)	103	60
108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI MGR	12008	(null)	101	100
109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI ACCOUNT	9000	(null)	108	100
114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU MAN	11000	(null)	100	30
120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST MAN	8000	(null)	100	50
121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST MAN	8200	(null)	100	50
122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST MAN	7900	(null)	100	50
123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST MAN	6500	(null)	100	50
124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST MAN	5800	(null)	100	50
137	Renske	Ladwig	RLADWIG	650.121.1234	14-JUL-03	ST CLERK	3600	(null)	123	50
141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-03	ST CLERK	3500	(null)	124	50
145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA MAN	14000	0.4	100	80
146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA MAN	13500	0.3	100	80
147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA MAN	12000	0.3	100	80
148	Gerald	Cambraut	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA MAN	11000	0.3	100	80
149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA MAN	10500	0.2	100	80
150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA REP	10000	0.3	145	80
151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA REP	9500	0.25	145	80
152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA REP	9000	0.25	145	80

Task 5

Worksheet Query Builder

```
SELECT first_name || ' ' || last_name AS employee_name, hire_date,
       ROUND((SYSDATE - hire_date)/365, 2) AS experience_years
FROM employees;
```

Query Result x

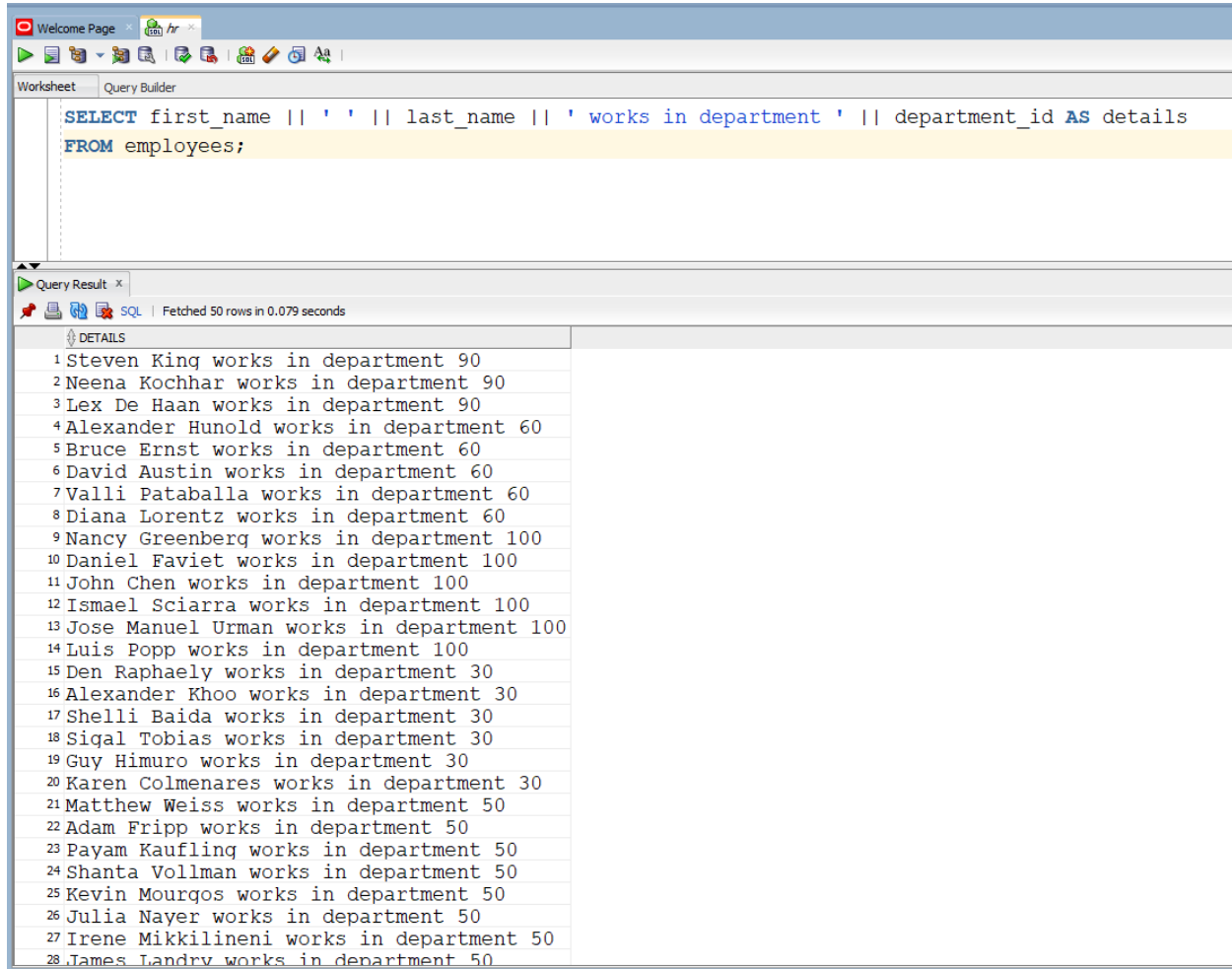
SQL | Fetched 50 rows in 0.004 seconds

	EMPLOYEE_NAME	HIRE_DATE	EXPERIENCE_YEARS
1	Steven King	17-JUN-03	22.67
2	Neena Kochhar	21-SEP-05	20.4
3	Lex De Haan	13-JAN-01	25.09
4	Alexander Hunold	03-JAN-06	20.12
5	Bruce Ernst	21-MAY-07	18.74
6	David Austin	25-JUN-05	20.64
7	Valli Pataballa	05-FEB-06	20.03
8	Diana Lorentz	07-FEB-07	19.02
9	Nancy Greenberg	17-AUG-02	23.5
10	Daniel Faviet	16-AUG-02	23.5
11	John Chen	28-SEP-05	20.38
12	Ismael Sciarra	30-SEP-05	20.38
13	Jose Manuel Urman	07-MAR-06	19.95
14	Luis Popp	07-DEC-07	18.19
15	Den Raphaely	07-DEC-02	23.19
16	Alexander Khoo	18-MAY-03	22.75
17	Shelli Baida	24-DEC-05	20.15
18	Sigal Tobias	24-JUL-05	20.56
19	Guy Himuro	15-NOV-06	19.25
20	Karen Colmenares	10-AUG-07	18.52
21	Matthew Weiss	18-JUL-04	21.58
22	Adam Fripp	10-APR-05	20.85
23	Payam Kaufling	01-MAY-03	22.8
24	Shanta Vollman	10-OCT-05	20.35
25	Kevin Mourgos	16-NOV-07	18.25

Task 6

Welcome Page											
Worksheet Query Builder											
<pre> SELECT * FROM (SELECT e.*, ROW_NUMBER() OVER (PARTITION BY department_id ORDER BY salary DESC) rn FROM employees e) WHERE rn <= 5; </pre>											
Query Result											
All Rows Fetched: 36 in 0.007 seconds											
EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	RN
1	200 Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD ASST	4400	(null)	101	10	1
2	201 Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK MAN	13000	(null)	100	20	1
3	202 Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK REP	6000	(null)	201	20	2
4	114 Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU MAN	11000	(null)	100	30	1
5	115 Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU CLERK	3100	(null)	114	30	2
6	116 Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU CLERK	2900	(null)	114	30	3
7	117 Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU CLERK	2800	(null)	114	30	4
8	118 Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU CLERK	2600	(null)	114	30	5
9	203 Susan	Mavris	SMAVRIS	515.123.7777	07-JUN-02	HR REP	6500	(null)	101	40	1
10	121 Adam	Frapp	AFRIPP	650.123.2234	10-APR-05	ST MAN	8200	(null)	100	50	1
11	120 Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST MAN	8000	(null)	100	50	2
12	122 Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST MAN	7900	(null)	100	50	3
13	123 Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST MAN	6500	(null)	100	50	4
14	124 Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST MAN	5800	(null)	100	50	5
15	103 Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT PROG	9000	(null)	102	60	1
16	104 Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT PROG	6000	(null)	103	60	2
17	105 David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT PROG	4800	(null)	103	60	3
18	106 Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT PROG	4800	(null)	103	60	4
19	107 Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT PROG	4200	(null)	103	60	5
20	204 Hermann	Baer	HBAER	515.123.8888	07-JUN-02	PR REP	10000	(null)	101	70	1
21	145 John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA MAN	14000	0.4	100	80	1
22	146 Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA MAN	13500	0.3	100	80	2
23	147 Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA MAN	12000	0.3	100	80	3
24	148 Ties	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA REP	11500	0.25	148	80	4

Task 7



The screenshot shows a database query tool interface. The top bar includes a 'Welcome Page' tab and a toolbar with various icons. Below the toolbar, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying a SQL query in a text area. The query is:

```
SELECT first_name || ' ' || last_name || ' works in department ' || department_id AS details
FROM employees;
```

Below the query editor, there is a 'Query Result' tab. It shows the results of the query, fetched in 0.079 seconds. The results are displayed in a table with a single column named 'DETAILS'. The table contains 28 rows of data, each showing an employee's name followed by their department ID.

DETAILS
1 Steven King works in department 90
2 Neena Kochhar works in department 90
3 Lex De Haan works in department 90
4 Alexander Hunold works in department 60
5 Bruce Ernst works in department 60
6 David Austin works in department 60
7 Valli Pataballa works in department 60
8 Diana Lorentz works in department 60
9 Nancy Greenberg works in department 100
10 Daniel Faviot works in department 100
11 John Chen works in department 100
12 Ismael Sciarra works in department 100
13 Jose Manuel Urman works in department 100
14 Luis Popp works in department 100
15 Den Raphaely works in department 30
16 Alexander Khoo works in department 30
17 Shelli Baida works in department 30
18 Sigal Tobias works in department 30
19 Guy Himuro works in department 30
20 Karen Colmenares works in department 30
21 Matthew Weiss works in department 50
22 Adam Fripp works in department 50
23 Payam Kaufling works in department 50
24 Shanta Vollman works in department 50
25 Kevin Mourgos works in department 50
26 Julia Nayer works in department 50
27 Irene Mikkilineni works in department 50
28 James Landry works in department 50

Task 8

The screenshot shows the SQL Developer interface with a query in the Query Builder. The query filters employees hired before January 1, 2015, and whose job is not a manager with a salary greater than 80,000. The results pane shows 12 rows of employee data.

```
SELECT *
FROM employees
WHERE hire_date < TO_DATE('01-JAN-2015','DD-MON-YYYY')
AND (
    (job_id LIKE '%MAN%' AND salary <= 80000)
    OR (job_id NOT LIKE '%MAN%' AND salary > 80000)
);
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU MAN	11000	(null)	100	30
2	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST MAN	8000	(null)	100	50
3	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST MAN	8200	(null)	100	50
4	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST MAN	7900	(null)	100	50
5	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST MAN	6500	(null)	100	50
6	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST MAN	5800	(null)	100	50
7	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA MAN	14000	0.4	100	80
8	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA MAN	13500	0.3	100	80
9	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA MAN	12000	0.3	100	80
10	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA MAN	11000	0.3	100	80
11	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA MAN	10500	0.2	100	80
12	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK MAN	13000	(null)	100	20

Task 9

The screenshot shows the SQL Developer interface with a query in the Query Builder. The query uses a subquery to filter employees whose department has an average salary greater than 70,000. The results pane shows the first two columns of the result set.

```
SELECT UPPER(first_name || ' ' || last_name) AS name_upper,
       LENGTH(first_name || last_name) AS name_length
FROM employees
WHERE department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING AVG(salary) > 70000
);
```

NAME_UP...	NAME_LE...
------------	------------

Task 10

The screenshot shows a database query tool interface. At the top, there's a 'Welcome Page' tab and a 'hr' database tab. Below the tabs is a toolbar with various icons. The main area is divided into two sections: 'Worksheet' and 'Query Builder'. The 'Query Builder' section contains the following SQL query:

```
SELECT first_name || ' ' || last_name AS employee_name,  
       salary + (salary * 0.10) AS total_income  
FROM employees;
```

Below the query editor, there's a 'Query Result' tab. It shows the results of the query, indicating that 50 rows were fetched in 0.004 seconds. The results are displayed in a table with two columns: 'EMPLOYEE_NAME' and 'TOTAL_INCOME'.

	EMPLOYEE_NAME	TOTAL_INCOME
1	Steven King	26400
2	Neena Kochhar	18700
3	Lex De Haan	18700
4	Alexander Hunold	9900
5	Bruce Ernst	6600
6	David Austin	5280
7	Valli Pataballa	5280
8	Diana Lorentz	4620
9	Nancy Greenberg	13208.8
10	Daniel Faviet	9900
11	John Chen	9020
12	Ismael Sciarra	8470
13	Jose Manuel Urman	8580
14	Luis Popp	7590
15	Den Raphaely	12100
16	Alexander Khoo	3410
17	Shelli Baida	3190
18	Sigal Tobias	3080
19	Guy Himuro	2860
20	Karen Colmenares	2750
21	Matthew Weiss	8800
22	Adam Fripp	9020
23	Payam Kaufling	8690
24	Shanta Vollman	7150
25	Kevin Mourgos	6380
26	Julia Nayer	3520
27	Irene Mikkilineni	2970

POST-LAB TASKS:

Task 11

Welcome Page		
Query Builder		
<pre>SELECT first_name ' ' last_name AS employee_name, salary, ROUND((salary / (SELECT SUM(salary) FROM employees)) * 100, 2) AS salary_percentage FROM employees;</pre>		
Query Result		
SQL Fetched 50 rows in 0.003 seconds		
EMPLOYEE_NAME	SALARY	SALARY_PERCENTAGE
1 Steven King	24000	3.47
2 Neena Kochhar	17000	2.46
3 Lex De Haan	17000	2.46
4 Alexander Hunold	9000	1.3
5 Bruce Ernst	6000	0.87
6 David Austin	4800	0.69
7 Valli Pataballa	4800	0.69
8 Diana Lorentz	4200	0.61
9 Nancy Greenberg	12008	1.74
10 Daniel Faviet	9000	1.3
11 John Chen	8200	1.19
12 Ismael Sciarra	7700	1.11
13 Jose Manuel Urman	7800	1.13
14 Luis Popp	6900	1
15 Den Raphaely	11000	1.59
16 Alexander Khoo	3100	0.45
17 Shelli Baida	2900	0.42
18 Sigal Tobias	2800	0.4
19 Guy Himuro	2600	0.38
20 Karen Colmenares	2500	0.36
21 Matthew Weiss	8000	1.16
22 Adam Fripp	8200	1.19
23 Payam Kaufling	7900	1.14
24 Shanta Vollman	6500	0.94
25 Kevin Mourgos	5800	0.84
26 Julia Nayer	3200	0.46
27 Tero Mikkilineni	2700	0.39

Task 12

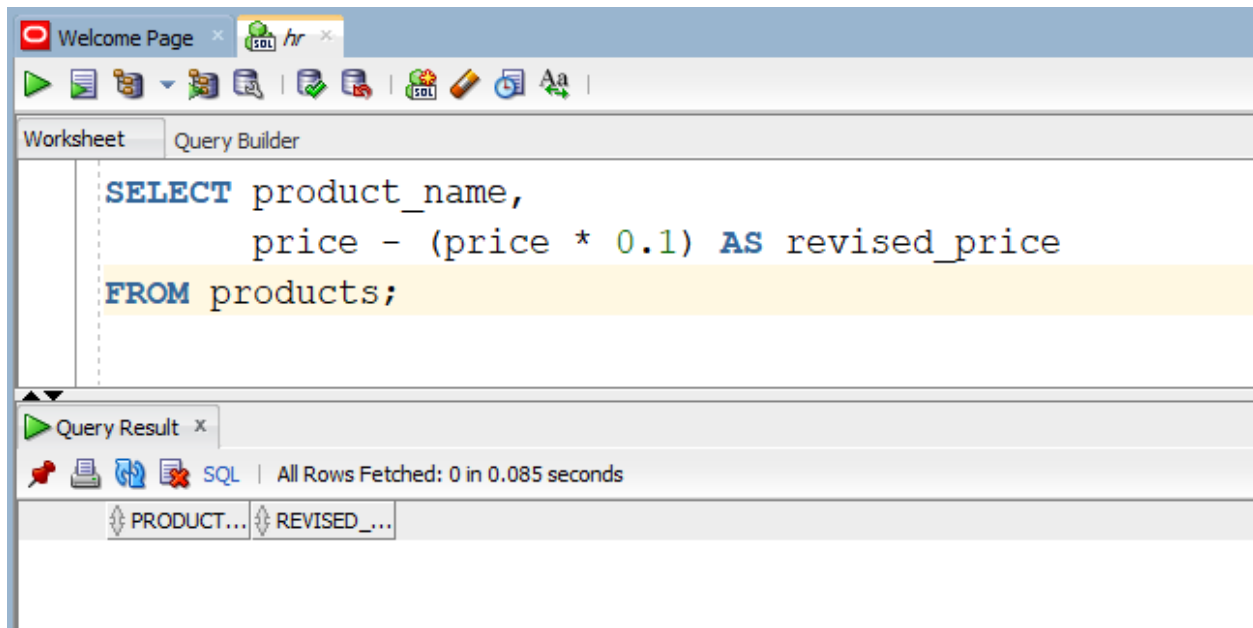
The screenshot shows a SQL development environment with two main panes. The top pane, titled 'Query Builder', contains the following SQL query:

```
SELECT UPPER(SUBSTR(d.department_name, 1, 3)) AS dept_short
FROM departments d;
```

The bottom pane, titled 'Query Result', displays the results of the query. It shows a table with one column, 'DEPT_SHORT', and 27 rows of data. The results are as follows:

DEPT_SHORT
1 ADM
2 MAR
3 PUR
4 HUM
5 SHI
6 IT
7 PUB
8 SAL
9 EXE
10 FIN
11 ACC
12 TRE
13 COR
14 CON
15 SHA
16 BEN
17 MAN
18 CON
19 CON
20 OPE
21 IT
22 NOC
23 IT
24 GOV
25 RET
26 REC
27 PAY

Task 13



Task 14

[illegible]

Task 15

The screenshot displays the Oracle SQL Developer interface. The top toolbar includes icons for running queries, saving, and other standard database operations. The 'Query Builder' tab is active, showing the following SQL query:

```
SELECT first_name || ' ' || last_name AS employee_name,  
       NVL(TO_CHAR(commission_pct), 'No Commission') AS commission  
FROM employees;
```

Below the query editor, the 'Query Result' window shows the output of the query. It indicates that 50 rows were fetched in 0.005 seconds. The results are displayed in a table with two columns: EMPLOYEE_NAME and COMMISSION.

EMPLOYEE_NAME	COMMISSION
1 Steven King	No Commission
2 Neena Kochhar	No Commission
3 Lex De Haan	No Commission
4 Alexander Hunold	No Commission
5 Bruce Ernst	No Commission
6 David Austin	No Commission
7 Valli Pataballa	No Commission
8 Diana Lorentz	No Commission
9 Nancy Greenberg	No Commission
10 Daniel Faviat	No Commission
11 John Chen	No Commission
12 Ismael Sciarra	No Commission
13 Jose Manuel Urman	No Commission
14 Luis Popp	No Commission
15 Den Raphaely	No Commission
16 Alexander Khoo	No Commission
17 Shelli Baida	No Commission
18 Sigal Tobias	No Commission
19 Guy Himuro	No Commission
20 Karen Colmenares	No Commission
21 Matthew Weiss	No Commission
22 Adam Fripp	No Commission
23 Payam Kaufling	No Commission
24 Shanta Vollman	No Commission
25 Kevin Mourgos	No Commission
26 Julia Nayer	No Commission
27 Irene Mikkilineni	No Commission

Task 16

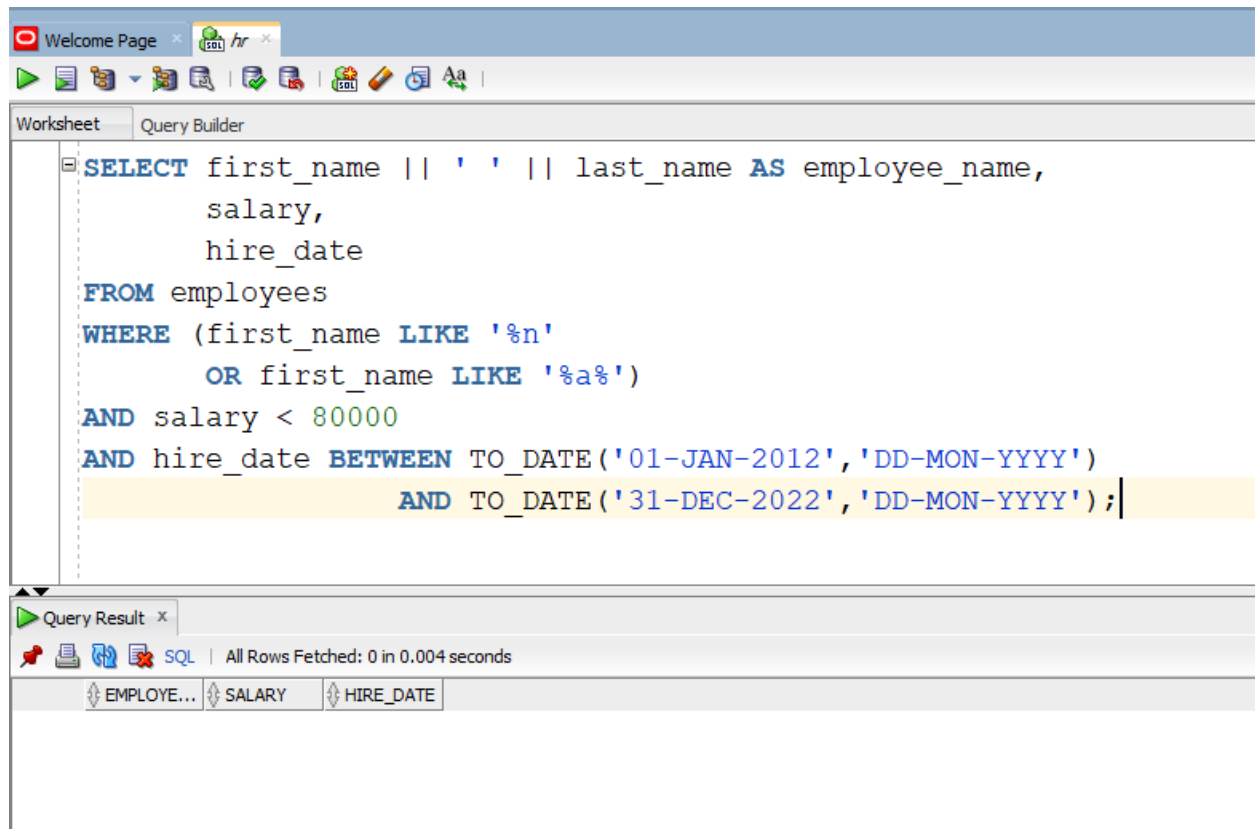
The screenshot shows a database query tool interface. At the top, there are tabs for 'Welcome Page' and 'hr'. Below the tabs is a toolbar with various icons. The main area is divided into two sections: 'Worksheet' and 'Query Builder'. The 'Query Builder' section contains the following SQL query:

```
SELECT UPPER(first_name || ' ' || last_name) AS name_upper,  
       SUBSTR(job_id, 1, 4) AS job_short,  
       TO_CHAR(hire_date, 'DD-MON-YYYY') AS formatted_date  
FROM employees;
```

Below the query, the 'Query Result' section shows the results of the query. It indicates that 50 rows were fetched in 0.007 seconds. The results are displayed in a table with three columns: NAME_UPPER, JOB_SHORT, and FORMATTED_DATE. The table contains 27 rows of data, numbered 1 through 27.

	NAME_UPPER	JOB_SHORT	FORMATTED_DATE
1	STEVEN KING	AD P	17-JUN-2003
2	NEENA KOCHHAR	AD V	21-SEP-2005
3	LEX DE HAAN	AD V	13-JAN-2001
4	ALEXANDER HUNOLD	IT P	03-JAN-2006
5	BRUCE ERNST	IT P	21-MAY-2007
6	DAVID AUSTIN	IT P	25-JUN-2005
7	VALLI PATABALLA	IT P	05-FEB-2006
8	DIANA LORENTZ	IT P	07-FEB-2007
9	NANCY GREENBERG	FI M	17-AUG-2002
10	DANIEL FAVIET	FI A	16-AUG-2002
11	JOHN CHEN	FI A	28-SEP-2005
12	ISMAEL SCIARRA	FI A	30-SEP-2005
13	JOSE MANUEL URMAN	FI A	07-MAR-2006
14	LUIS POPP	FI A	07-DEC-2007
15	DEN RAPHAELY	PU M	07-DEC-2002
16	ALEXANDER KHOO	PU C	18-MAY-2003
17	SHELLI BAIDA	PU C	24-DEC-2005
18	SIGAL TOBIAS	PU C	24-JUL-2005
19	GUY HIMURO	PU C	15-NOV-2006
20	KAREN COLMENARES	PU C	10-AUG-2007
21	MATTHEW WEISS	ST M	18-JUL-2004
22	ADAM FRIPP	ST M	10-APR-2005
23	PAYAM KAUFING	ST M	01-MAY-2003
24	SHANTA VOLLMAN	ST M	10-OCT-2005
25	KEVIN MOURGOS	ST M	16-NOV-2007
26	JULIA NAYER	ST C	16-JUL-2005
27	IRENE MIKKILINENI	ST C	28-SEP-2006

Task 17



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running queries, saving, and editing. The 'Query Builder' tab is active, showing a SQL query that selects employee details based on specific criteria. The query is as follows:

```
SELECT first_name || ' ' || last_name AS employee_name,  
       salary,  
       hire_date  
FROM employees  
WHERE (first_name LIKE '%n'  
       OR first_name LIKE '%a%')  
AND salary < 80000  
AND hire_date BETWEEN TO_DATE('01-JAN-2012', 'DD-MON-YYYY')  
                  AND TO_DATE('31-DEC-2022', 'DD-MON-YYYY');
```

Below the query editor, the 'Query Result' window is visible, indicating that all rows were fetched successfully in 0.004 seconds. The result set columns are listed as EMPLOYEE..., SALARY, and HIRE_DATE.

Task 18

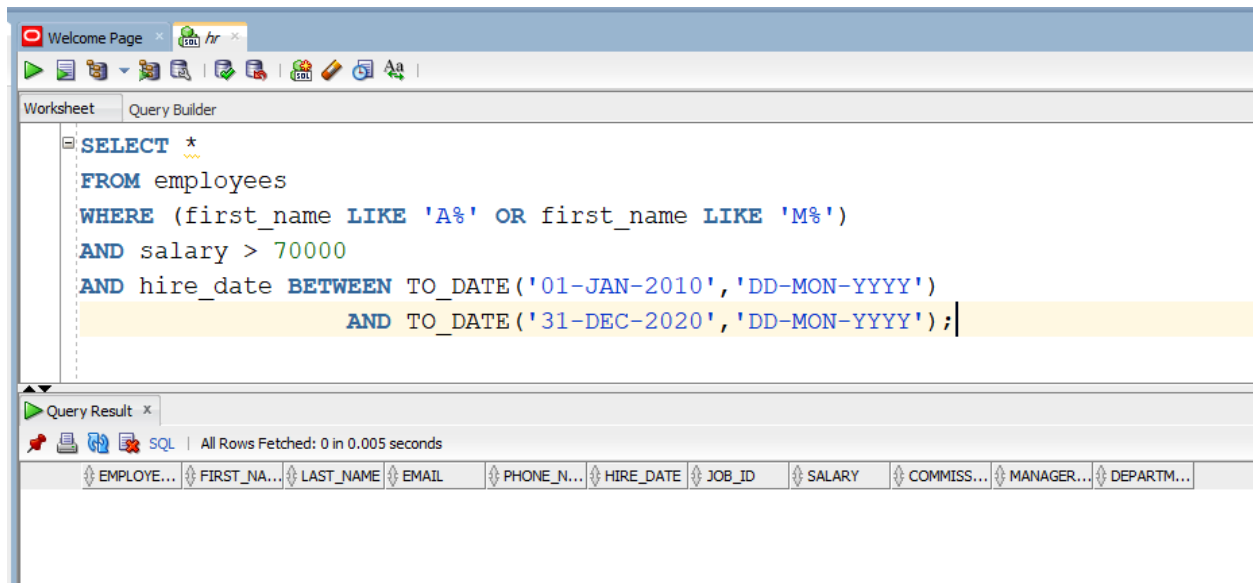
The screenshot shows a SQL query editor window with a 'Query Builder' tab. The query is as follows:

```
SELECT first_name || ' ' || last_name AS employee_name,  
       salary,  
       salary * 12 AS annual_income  
FROM employees;
```

Below the query editor, the 'Query Result' tab is active, displaying the results of the query. It indicates that 50 rows were fetched in 0.078 seconds. The results are shown in a table with three columns: EMPLOYEE_NAME, SALARY, and ANNUAL_INCOME.

	EMPLOYEE_NAME	SALARY	ANNUAL_INCOME
1	Steven King	24000	288000
2	Neena Kochhar	17000	204000
3	Lex De Haan	17000	204000
4	Alexander Hunold	9000	108000
5	Bruce Ernst	6000	72000
6	David Austin	4800	57600
7	Valli Pataballa	4800	57600
8	Diana Lorentz	4200	50400
9	Nancy Greenberg	12008	144096
10	Daniel Faviet	9000	108000
11	John Chen	8200	98400
12	Ismael Sciarra	7700	92400
13	Jose Manuel Urman	7800	93600
14	Luis Popp	6900	82800
15	Den Raphaely	11000	132000
16	Alexander Khoo	3100	37200
17	Shelli Baida	2900	34800
18	Sigal Tobias	2800	33600
19	Guy Himuro	2600	31200
20	Karen Colmenares	2500	30000
21	Matthew Weiss	8000	96000
22	Adam Fripp	8200	98400
23	Payam Kaufling	7900	94800
24	Shanta Vollman	6500	78000
25	Kevin Mourgos	5800	69600
26	Julia Nayer	3200	38400
27	Irene Mikkilineni	2700	32400
28	James Landry	2400	28800

Task 19

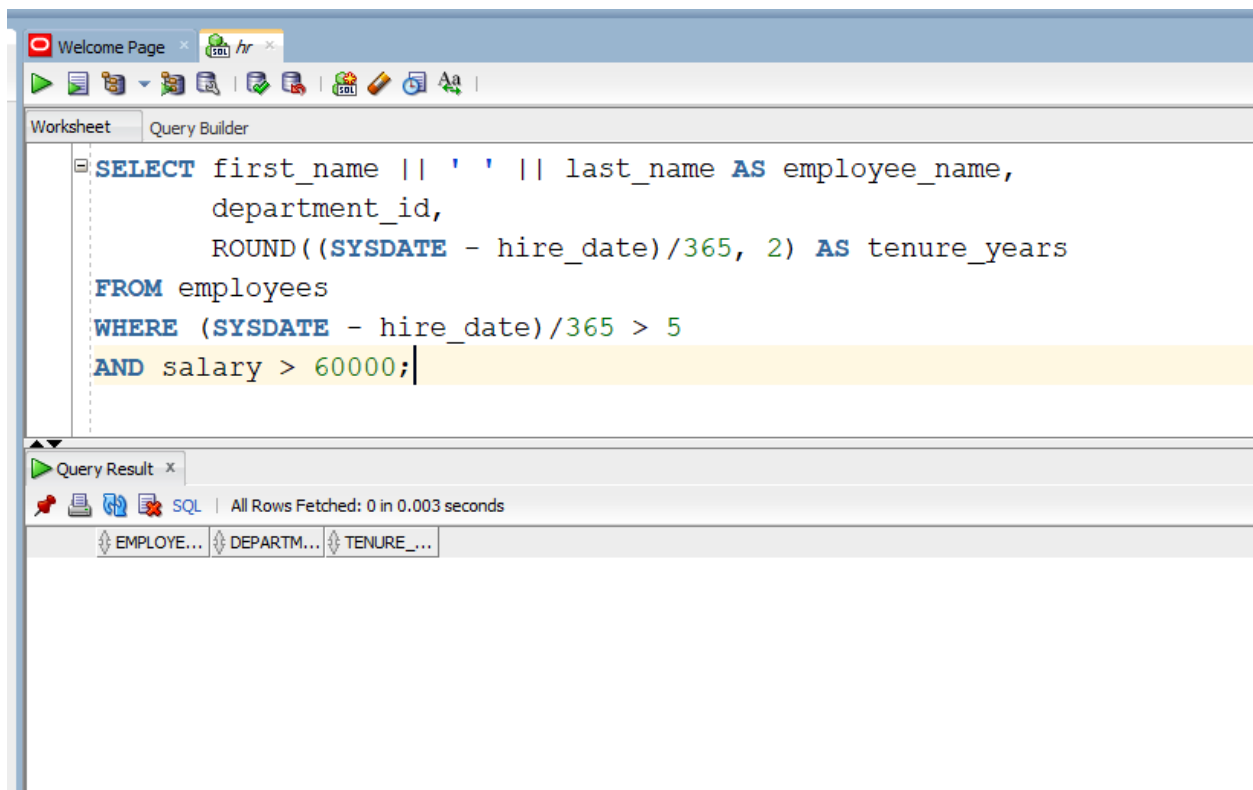


The screenshot shows the SQL Developer interface with the Query Builder tab active. The query is as follows:

```
SELECT *
FROM employees
WHERE (first_name LIKE 'A%' OR first_name LIKE 'M%')
AND salary > 70000
AND hire_date BETWEEN TO_DATE('01-JAN-2010','DD-MON-YYYY')
AND TO_DATE('31-DEC-2020','DD-MON-YYYY');
```

The Query Result pane shows "All Rows Fetched: 0 in 0.005 seconds". The column list at the bottom includes: EMPLOYEE..., FIRST_NA..., LAST_NAME, EMAIL, PHONE_N..., HIRE_DATE, JOB_ID, SALARY, COMMISS..., MANAGER..., and DEPARTM...

Task 20



The screenshot shows the SQL Developer interface with the Query Builder tab active. The query is as follows:

```
SELECT first_name || ' ' || last_name AS employee_name,
       department_id,
       ROUND((SYSDATE - hire_date)/365, 2) AS tenure_years
FROM employees
WHERE (SYSDATE - hire_date)/365 > 5
AND salary > 60000;
```

The Query Result pane shows "All Rows Fetched: 0 in 0.003 seconds". The column list at the bottom includes: EMPLOYEE..., DEPARTM..., and TENURE_...